

SUJITH MAKAM

Software Engineer — Cloud & AI

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📍 Bengaluru, India

Summary

Software Engineer with 1+ year of experience building production systems across AWS and Azure, with hands-on expertise in cloud infrastructure, Infrastructure-as-Code, disaster recovery, automation, observability, and distributed systems. Experienced in AWS CDK, serverless architectures, and AI/LLM platforms; author of 4 IEEE-published papers in ML and network security.

Education

Indian Institute of Information Technology (IIIT), Dharwad

DEC 2021 – JULY 2025

B.Tech in Computer Science and Engineering

CGPA: 8.41

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, C, SQL

Cloud & Infrastructure: AWS (Lambda, Bedrock, S3, DynamoDB, EventBridge, IAM, KMS, CDK, Amazon Connect), Azure (Functions, Cosmos DB), GCP (Cloud Run, Vertex AI), Docker, CI/CD, Infrastructure-as-Code

SRE & DevOps: Reliability Engineering, Disaster Recovery, Failure Recovery, Automation, Monitoring, Alerting, OpenTelemetry, CloudWatch, Linux, Git

Backend & Distributed Systems: Node.js, Express.js, REST APIs, WebSockets, JSON-RPC, Microservices, Serverless, Distributed Systems

Databases: MongoDB, MySQL, PostgreSQL, DynamoDB, Cosmos DB, Vector Databases, SQL Optimization

AI & Machine Learning: Generative AI, Agentic AI, LLMs, RAG, Multi-Agent Systems, LLM Evaluation, Prompt Engineering, MCP, A2A Protocol, Google ADK, CrewAI, Azure AI Foundry, MLOps

Tools & Platforms: Langfuse, AgentOps, Claude, Copilot, React.js, Pytest, Cognigy, Genesys

Experience

CGI

Jul. 2025 – Present

Software Engineer

- Designed and shipped enterprise **agentic-AI** solutions across Azure AI Foundry, Cognigy, Genesys and AWS, automating customer-support workflows and improving operational efficiency.
- Built production LLM applications (call summarization, automated QA evaluation) with Azure OpenAI and **RAG** over enterprise knowledge bases.
- Developed multi-agent workflows for autonomous task execution, reducing manual intervention and improving SLA compliance.
- Provisioned complete **Amazon Connect** environments as **Infrastructure-as-Code** with **AWS CDK (TypeScript)**, enabling automated multi-region contact-center deployments.

Indian Institute of Science, Bangalore

May. 2024 – July. 2024

Research Intern

- Investigated LLM Pruning techniques for improving model efficiency, optimizing the attention and feed-forward layers of GPT-2. Achieved a 5.2% reduction in instruction cache misses through tiled matrix multiplication implementation.
- Conducted detailed profiling of GPT-2 using Intel VTune, identifying bottlenecks and enhancing cache efficiency, which resulted in a 4x speedup in attention layer execution.
- Optimized identified bottlenecks in the attention and feed-forward layers of GPT-2 and Reduced memory bandwidth usage by 34.8 %, decreasing memory-bound issues and ensuring more efficient execution
- Implemented custom classes (e.g., tiled matrix multiplication) for performance improvement in the attention layer.

Projects

LegacyLift | Multi-Agent, MCP, A2A, Next.js



- Built a multi-agent system making legacy **COBOL** logic callable by AI agents without rewriting the mainframe; **top 13 of 200+ entries at CGI Envision 2026 Hackathon**.
- Implemented 5 collaborating agents over an **A2A** message bus with a shared context graph, parsing COBOL source to expose routines and auto-generate **MCP** tools.

- Streamed live execution state to a Next.js UI over a WebSocket relay behind a provider abstraction to avoid vendor SDK coupling.

AI Governance Platform | *Azure AI Foundry, Google ADK, CrewAI*



- Built an AI governance platform for multi-agent workflows using Azure AI Foundry, Google ADK, CrewAI and Agent Gateway for secure LLM-agent orchestration.
- Integrated Langfuse and AgentOps for agent observability, prompt tracing, KPI monitoring and latency analysis.
- Implemented Responsible-AI controls: prompt-injection prevention, sensitive-data masking, API isolation and policy enforcement.

Amazon Connect Disaster Recovery & Reliability Framework | *AWS CDK (IaC), Step Functions, Lambda*



- Engineered a production-grade Disaster Recovery synchronization solution for Amazon Connect using Boto3, AWS SDK, Lambda, Step Functions, EventBridge and SNS to orchestrate automated **cross-region configuration synchronization and failover**.
- Provisioned the entire DR stack as **Infrastructure-as-Code** with **AWS CDK (TypeScript)**, deploying repeatable, version-controlled cross-region environments to support reliable failover and rapid recovery.
- Scheduled synchronization runs during non-business hours and implemented automated **failure detection, retries and SNS-driven alerting**, reducing recovery time and manual intervention during **failover**.
- Deployed to production with a **99.9%** successful-run rate, eliminating manual DR configuration drift and improving **failover readiness, high availability and business continuity** for enterprise contact-center operations.

Certifications

CGI: Inside Agentic AI (Foundations; Core Architecture), AI Guardrails & Governance, Elements of an AI Architect, Reusable AI Architecture Patterns, Cloud Computing & MLOps, Microsoft Foundry for Enterprise AI
Udemy: Full Stack Web Development, Machine Learning A-Z

Research Publications

- FAPL-DM-BC: Adaptive Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for the IoVs – **IEEE COMSNETS 2026**
- Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI – **IEEE FNWF 2025**
- FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT – **IEEE FNWF 2025**
- Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars – **IEEE SPACE 2025**

Achievements

- **CGI Envision 2026 Hackathon:** Top 13 of 200+ submissions (LegacyLift).
- **Andhra Pradesh State Board Topper:** 990/1000.